CS-302: Artificial Intelligence

LECTURE NO 14:

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Organization of Lecture

1. Supervised, Unsupervised and Reinforcement Learning.
2. Recurrent Neural Network (RNN).
3. Natural Language Processing (NLP).
4. Expert System Architecture.
5. Constraint Satisfaction.
6. Effect of Learning Rate (LR) on Back Propagation Algorithm.

1. Supervised, Unsupervised and Reinforcement Learning (1/4)

Supervised Learning: Happens in presence of Supervisor.

↳ output is already known.

↳ machine is fed with lots of input data.

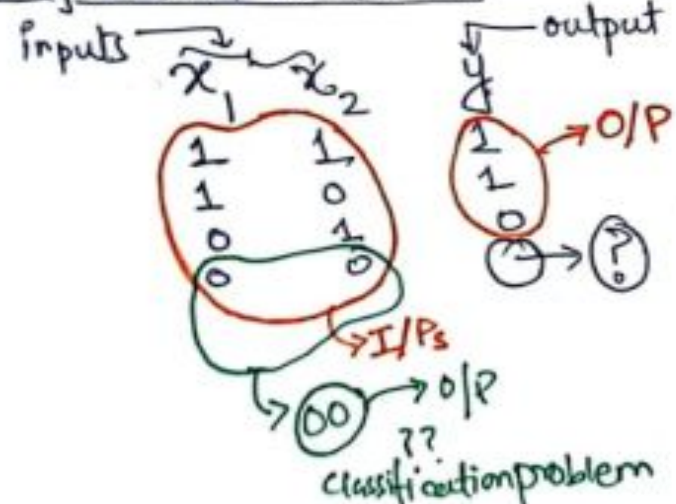
↳ Model is built to predict the outcomes.

[Input ⟷ output] } Labelled DataSet.

↳ Fast Learning mechanism with high accuracy.

↳ includes Prediction (Regression) and classification Problems.

↳ Classification Problem



1. Supervised, Unsupervised and Reinforcement Learning (2/4)

↳ Examples: (Supervised Learning)
- SVM (Support Vector Machine)
- KNN (K Nearest Neighbour)
- ANN (Artificial Neural Network)
- Decision Trees & so on.

Unsupervised Learning: Happens without help of supervisor

↳ Independent Learning process.

↳ no output mapping with input.

↳ Unlabelled Dataset.

↳ Detect Hidden Patterns

↳ Commonly used unsupervised learning techniques:

↳ clustering & Association Rule mining.

clustering ⇓ K-means clustering

Association Rule mining ⇓ Apriori

1. Supervised, Unsupervised and Reinforcement Learning (3/4)

Reinforcement Learning: Learns from Feedback and past experiences.

- Long term iterative process
- More feed back ⇒ More Accurate System
- Also called Markov Decision Process.
- E.g:- Robots Training, Self Driving Cars etc.
  - Deep Adversial N/W, Q Learning.
    - Quality

1. Supervised, Unsupervised and Reinforcement Learning (4/4)

Supervised Learning Algorithm	Unsupervised Learning Algorithm
output is known for every input.	output is unknown.
Learn from labelled data set.	It finds hidden patterns or associations among data items.
It is used to predict outcome of new value (predictive in nature)	Descriptive in nature.
Classification, Regression	Clustering, Association Algo:
Examples: Linear Regression, KNN, SVM etc.	Examples: K-means, Apriori etc.
It is more Accurate	It is less Accurate

2. Recurrent Neural Network (RNN) (1/4)

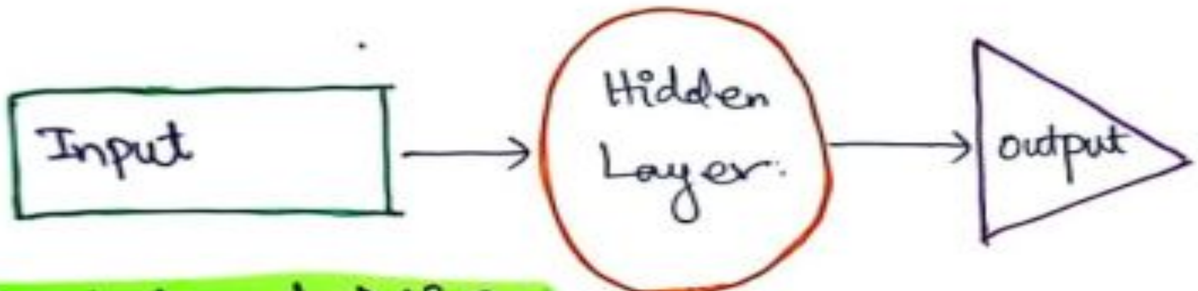
In RNN output from previous steps are fed as input to the current step (In tradition I/P & O/P are independent).

↳ useful when it is required to predict next word of sentence.

↳ HIDDEN LAYER is used.

↳ It remembers some information about previous sequence [Hidden State]

↳ Hidden Layer remembers information about sequence.



RNN Characteristics–

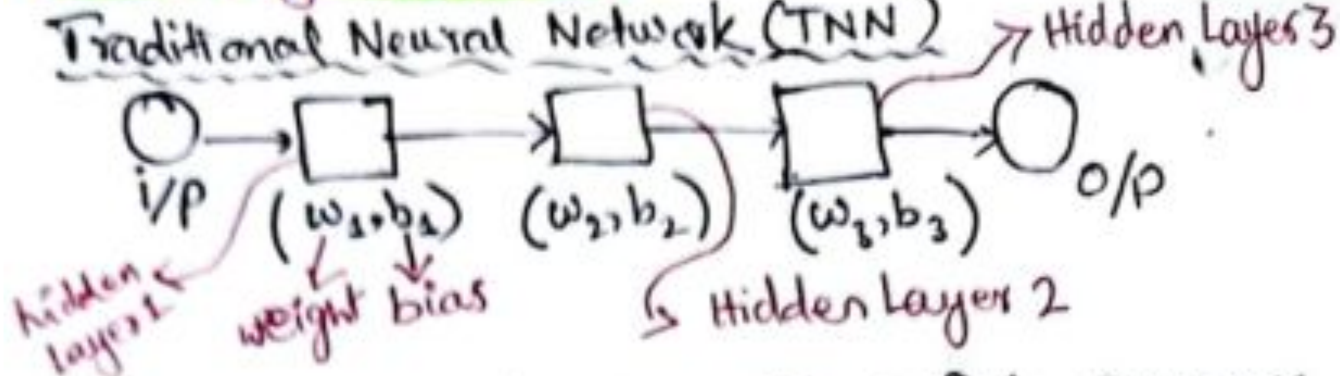
↳ It has a memory that remembers all the information about what has been calculated.

↳ It reduces the complexity of parameters because of the above characteristic.

2. Recurrent Neural Network (RNN) (2/4)

Working of RNN:

Traditional Neural Network (TNN)



→ We can convert above TNN into RNN as:

↳ Merge all (3) hidden layers of TNN into one / single hidden Layer:

$$h_t = f(h_{t-1}, x_t)$$

2. Recurrent Neural Network (RNN) (3/4)

Where:

h_t = current state which we want to find out

h_{t-1} = Previous state

x_t = Input State

Formula for applying the Activation Function:

$$h_t = \tanh(W_{hh} h_{t-1} + W_{xh} x_t)$$

where: W_{hh} = weight at recurrent neuron

W_{xh} = weight at input neuron.

Calculating Output:

$$\underbrace{Y_t}_{\text{O/P}} = \underbrace{W_{hy}}_{\text{Weight at the output layer.}} \overbrace{h_t}^{\text{Current State}}$$

2. Recurrent Neural Network (RNN) (4/4)

Advantages of RNNs

i) Useful in Times Series Prediction.

Because it remembers information through time. Example is LSTM (Long Short Term Memory).

ii) Used to extend pixel neighbourhood

When used with Convolutional Layer.

Disadvantages of RNNs

i) Problems like Gradient Vanishing

ii) Training is difficult.

Because Learning from previous states

iii) Not very efficient in long sequences processing. (Time Consuming)

3. Natural Language Processing (NLP) (1/3)

NLP is a branch of AI that helps computers to understand, interpret and manipulate human language.

↳ Following tasks that can be done are:

- ↳ ① Translation (Google Translator)
- ↳ ② Summarization
- ↳ ③ Speech Recognition (Latest TV Remote Control)
- ↳ ④ Sentimental Analysis

3. Natural Language Processing (NLP) (2/3)

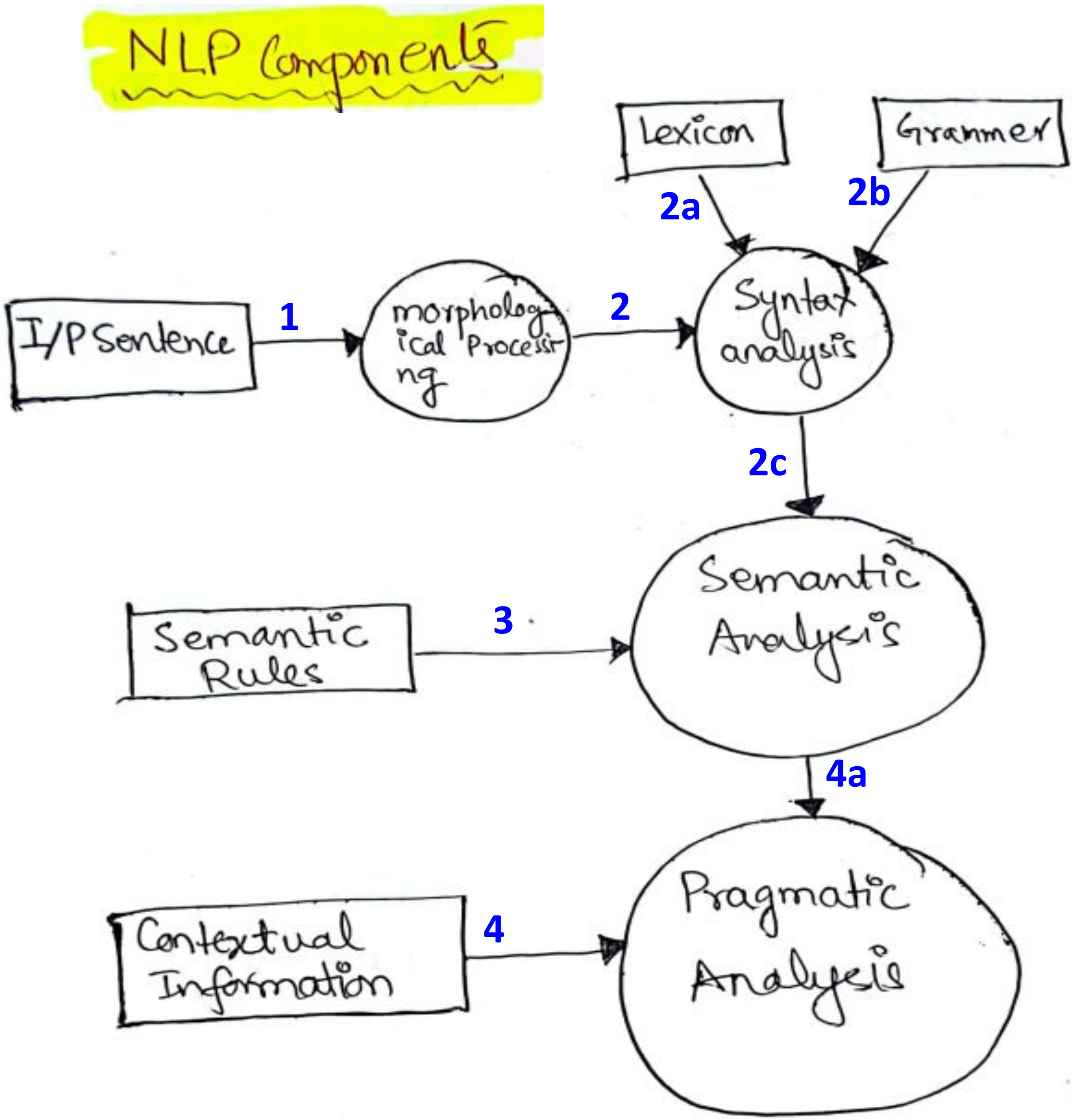
Morphological Processing: Vocabulary that includes words & expressions. It depicts word structure information. It includes division of text into paragraphs, words, sentence.

Syntax Analysis: Rules, principles that governs the sentence structure of individual language.

Semantic Analysis: Assigns meaning. It shows how words are associated with each other.

Pragmatic Analysis: Overall communicative and social context and its effects on interpretation ⇒ It derives meaningful use.

3. Natural Language Processing (NLP) (3/3)



4. Expert System Architecture (1/2)

Expert system in AI are Computer Application with the level of human intelligence and expertise.

It draws upon knowledge of human experts captured in a knowledge base to solve problems that require human expertise.

Characteristics:

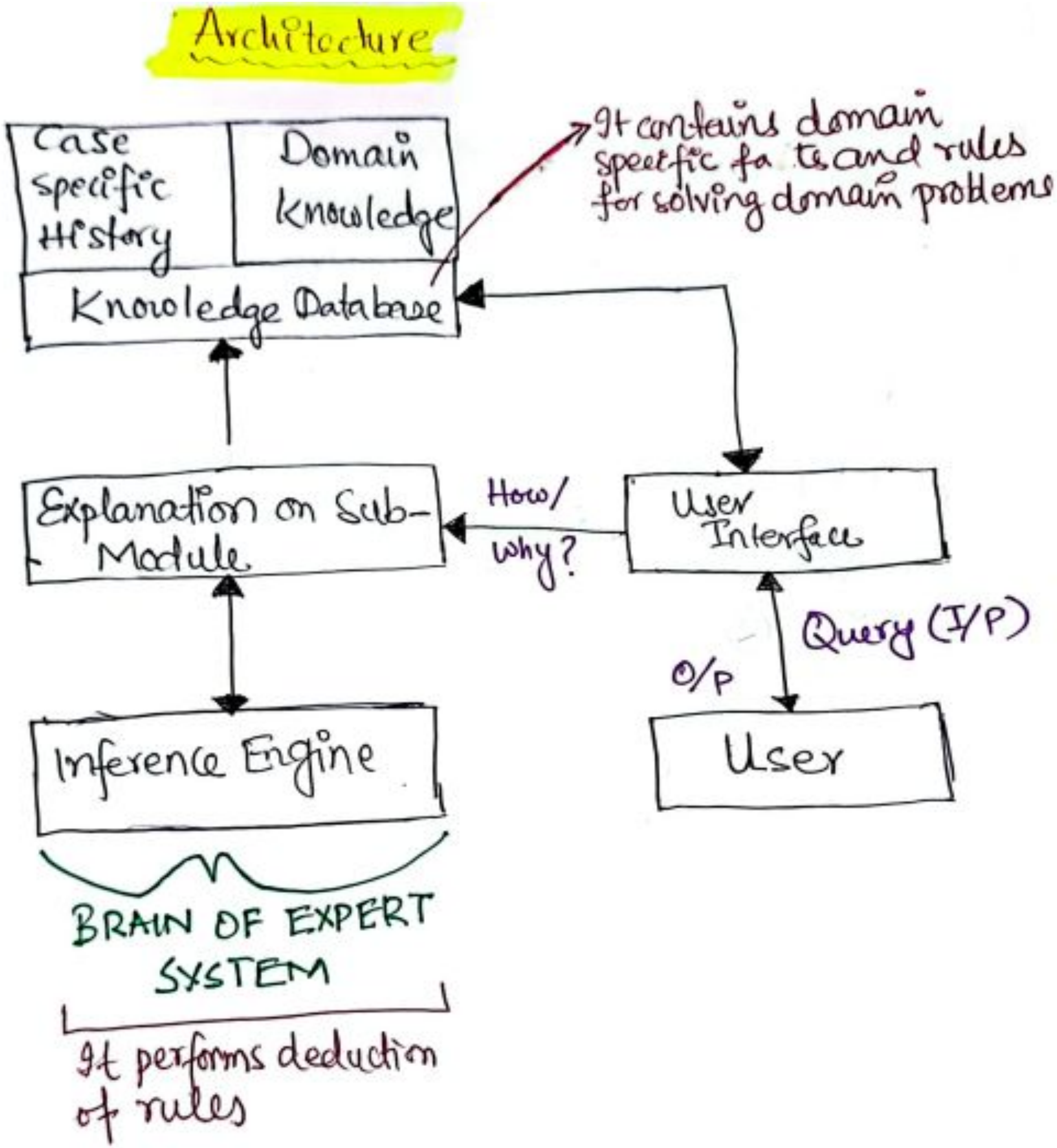
i) High Performance

ii) Reliable

iii) Highly Responsive

iv) Under standable

4. Expert System Architecture (2/2)



5. Constraint Satisfaction (1/4)

↳ It is a search procedure that operates in a space of constraint sets.

↳ Constraint Satisfaction Problems in AI have goal of discovering some problem state that satisfies a given set of constraints.

Process

↳ i) Constraints are discovered and propogated throughout the system.

ii) If still there is no solution, search begins.

↳ A Guess is made about something and added as a new constraint.

5. Constraint Satisfaction (2/4)

Constraint Satisfaction Probelm (CSP)

- → V [Variables Set] $\{V_1, V_2, \ldots V_n\}$
- → D [Domain] $\{D_1, D_2, \ldots D_n\}$
  - ↳ One domain for each variable
- → C [constraints].
  - ↳ specify allowable combination of values.

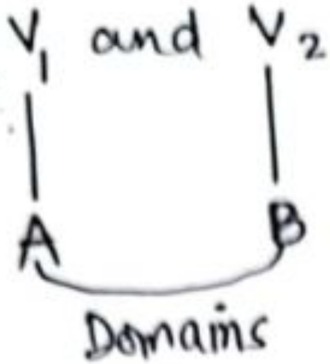
A Constraint can be represented as:

$$C_i = (\text{Scope}, \text{Relation})$$

- Scope → Set of variables that participate in Constraint
- Relation ↳ It defines values that a variable can take

5. Constraint Satisfaction (3/4)

→ Suppose we have two **variables** , V_1 & V_2 :



Constraint ①: values of V_1 and V_2 can't be same.

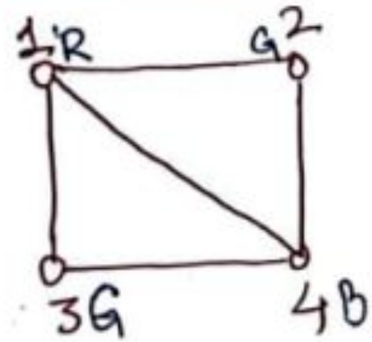
$$C_1 = \{ \underbrace{(V_1, V_2)}_{\text{Scope}},\ \underbrace{V_1 \neq V_2}_{\text{Relation}} \}$$

→ Constraint Satisfaction probelm (CSP) can be solved using Intelligent back tracking method.

→ Intelligent Back Tracking is used to only back track where conflict occurs.

5. Constraint Satisfaction (4/4)

NODE COLORING: It is a classical example of CSP.



V = {1, 2, 3, 4}

D = {Red, Green, Blue}

C = {adjacent nodes should not have same color}.

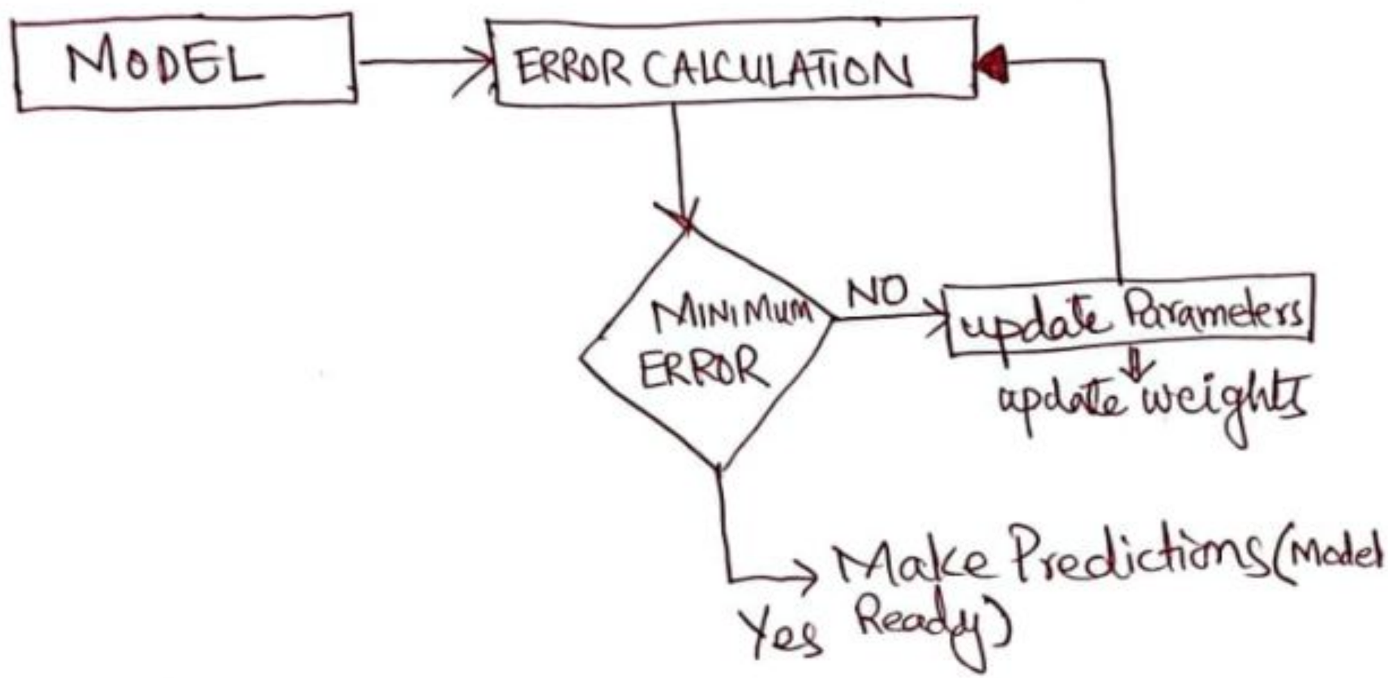
Four different Nodes:

	1	2	3	4
Initial	R, G, B	R, G, B	R, G, B	R, G, B
1=R	R	G, B	G, B	G, B
2=G	R	G	G, B	B
3=B	R	G	B	ERROR
3=G	R	G	G	B

Back tracking

6. Effect of Learning Rate (LR) on Back Propagation Algorithm (1/3)

- → Back propagation is supervised Learning algorithm for training multilayer perceptrons (ANN).
- → It looks for minimum value of error function using technique called Gradient Descent / Delta Rule.
- → Weights that minimized error function is then consid ered to be solution to Learning problem.



6. Effect of Learning Rate (LR) on Back Propagation Algorithm (2/3)

Learning Rate:

↳ It is the amount that the weights are updated during training. Also called stepsize & generally its range is [0.0 to 1.0]

↳ Weights are scaled by Learning Rate.

for example: Suppose previously the value of weight was 0.3 & found error 0.2. i.e W=0.3 & E=0.2.
Now the updation process can be written as:

$$0.3 + (0.2)(0.1) = 0.3 + 0.02$$

6. Effect of Learning Rate (LR) on Back Propagation Algorithm (3/3)

Effect of Learning Rate:

↳ LR controls rate or speed at which your model learns.

Large LR

→ Fast model Learning.

→ It may arrive at suboptimal weight.

→ Gradient Descent can increase the error when LR is High.

Small LR

→ Slow & optimal model learning but very large time.

→ It can stuck permanently with High training Error.

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Textbooks

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7. *Related documents from open source, mainly internet.*